

NYUTABARU AB, Japan

WMO: 478540

Lat: 32.084N

Lon: 131.451E

Elev: 79

StdP: 100.38

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.1	0.1	-9.8	1.6	3.9	-8.1	1.9	4.2	11.9	8.6	10.8	9.0	3.2	270	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.2	32.9	25.7	31.2	25.9	30.2	25.7	27.3	29.8	26.8	29.6	26.2	29.0	5.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	22.9	28.7	26.1	21.7	28.3	25.8	21.2	28.1	87.3	29.7	84.6	30.4	82.2	29.1	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.8	8.1	6.8	-4.0	35.4	1.1	1.3	-4.8	36.3	-5.5	37.0	-6.1	37.7	-6.9	38.6		
			-5.6	28.7	1.1	1.0	-6.3	29.4	-6.9	30.0	-7.5	30.6	-8.3	31.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.4	7.2	8.5	11.4	15.8	19.8	22.6	26.8	27.3	24.6	20.0	14.7	9.4
	DBStd	7.41	3.00	3.49	3.20	2.85	2.00	2.27	1.86	1.45	2.16	2.69	3.34	3.33
	HDD10.0	239	96	66	22	1	0	0	0	0	0	0	3	52
	HDD18.3	1335	344	276	216	84	8	0	0	0	0	14	116	277
	CDD10.0	2933	10	23	64	174	302	378	519	536	439	309	143	33
	CDD18.3	988	0	0	1	7	53	129	261	277	189	65	6	0
	CDH23.3	7544	1	1	2	23	146	598	2465	2782	1305	214	8	0
	CDH26.7	2099	0	0	0	3	16	112	800	887	265	17	1	0
Wind	WSAvg	3.1	3.4	3.3	3.4	3.2	2.9	2.8	3.1	3.1	2.9	2.9	2.8	3.2
Precipitation	PrecAvg	2468	68	97	156	192	227	494	360	264	306	160	102	67
	PrecMax	3952	172	180	408	342	497	860	967	548	604	468	256	157
	PrecMin	1557	1	38	47	57	76	166	74	42	60	13	30	2
	PrecStd	462	38	46	71	68	90	176	205	132	153	115	55	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.1	20.9	22.2	26.0	28.8	32.1	34.8	34.9	32.2	28.2	24.2	20.1
		MCWB	13.8	16.2	15.9	17.6	20.9	25.1	25.4	26.2	25.7	23.1	19.4	15.5
	2%	DB	15.9	18.1	20.1	23.2	26.1	29.8	32.9	32.9	30.1	26.9	22.8	18.0
		MCWB	12.0	14.5	14.9	17.6	19.8	25.2	25.6	25.9	25.4	22.3	18.3	14.0
	5%	DB	14.1	16.1	18.4	22.0	24.9	27.9	31.8	31.2	29.1	25.2	21.2	16.2
		MCWB	10.2	12.6	13.8	16.9	19.8	24.1	25.9	26.1	24.8	21.0	17.3	12.7
	10%	DB	12.9	14.2	17.0	20.9	23.9	26.2	30.2	30.2	28.1	24.2	20.1	15.1
		MCWB	9.3	10.7	13.1	16.6	19.1	22.9	25.8	25.9	24.0	20.1	16.5	11.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.6	17.5	18.2	21.4	23.9	26.5	28.1	28.4	28.2	25.5	21.4	17.8
		MCDB	17.3	19.6	20.9	22.8	26.0	29.4	30.3	29.9	29.4	26.9	23.3	19.0
	2%	WB	13.2	15.5	16.5	20.1	22.6	25.7	27.2	27.4	26.5	24.0	19.7	15.5
		MCDB	14.6	17.6	18.5	21.5	24.2	28.5	30.0	29.8	28.7	25.7	21.2	16.9
	5%	WB	11.4	13.8	15.3	18.7	21.4	24.8	26.7	26.9	25.8	22.7	18.6	13.9
		MCDB	13.3	15.6	17.4	20.7	23.0	27.1	29.6	29.7	28.0	24.4	20.3	16.0
	10%	WB	9.7	11.8	13.9	17.5	20.5	23.9	26.1	26.4	25.2	21.3	17.3	12.3
		MCDB	12.2	13.7	16.2	19.7	22.3	25.9	29.1	29.1	27.2	23.4	19.4	14.7
Mean Daily Temperature Range	5% DB	MDBR	9.5	9.0	8.9	8.4	7.3	5.8	6.4	6.2	6.3	7.3	8.4	9.3
		MCDBR	10.5	10.2	10.5	10.1	9.2	7.5	8.5	7.7	7.1	7.9	8.5	9.7
		MCWBR	7.5	7.4	7.1	6.5	5.5	4.3	3.8	3.7	4.0	4.9	5.6	6.7
	5% WB	MCDBR	8.8	8.6	8.5	7.6	6.7	6.6	6.6	6.5	6.0	5.9	6.7	7.9
		MCWBR	7.6	7.1	7.0	5.9	4.9	4.1	3.7	3.6	4.0	4.3	5.2	6.4
Clear-Sky Solar Irradiance	taub	0.409	0.454	0.515	0.538	0.550	0.554	0.539	0.526	0.508	0.473	0.424	0.397	
	taud	2.144	2.050	1.894	1.891	1.931	2.000	2.081	2.106	2.121	2.134	2.194	2.200	
	Ebn at Noon	786	787	767	770	763	757	767	771	767	760	765	776	
	Edh at Noon	125	151	189	198	191	178	164	157	149	137	118	112	
All-Sky Solar Radiation	RadAvg	2.72	3.24	4.16	4.76	4.90	3.92	5.07	5.08	4.16	3.50	2.77	2.49	
	RadStd	0.21	0.28	0.28	0.51	0.56	0.65	0.61	0.40	0.48	0.41	0.29	0.19	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page